

Unified Sizing Through AI & App-Based Solutions: A Strategic Business Proposal for the Indian Apparel Market

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Abstract

The Indian apparel market faces a persistent challenge: inconsistent sizing across brands leads to high return rates, customer dissatisfaction, and significant environmental waste. This paper proposes a unified, AI-driven sizing solution integrated via B2B licensing and B2C freemium models, targeting major e-commerce platforms such as Myntra and Ajio. The solution leverages computer vision, augmented reality, and machine learning to deliver accurate, brand-agnostic body measurements. Financial projections indicate a path to profitability within 18–24 months, with revenue streams spanning B2B licensing fees, premium B2C subscriptions, and anonymized data monetization—all compliant with India’s Digital Personal Data Protection Act (DPDPA). This paper outlines the problem, proposed technology, business model, go-to-market strategy, financial projections, risk analysis, and competitive landscape.

1 Executive Summary

The Indian online apparel market, valued at over USD 30 billion and growing at 18% CAGR, is constrained by a fundamental sizing fragmentation problem. Sizes labelled ‘M’

or ‘32’ vary dramatically across brands, resulting in return rates as high as 30–40% for some players. This translates into billions of rupees in annual waste—logistical, financial, and environmental.

This paper proposes a unified sizing intelligence layer: an AI and app-based platform that provides body measurement and fit prediction across any brand’s size chart. The solution has three revenue legs:

- **B2B Licensing:** API access sold to Myntra, Ajio, and other fashion e-tailers; brands embed our sizing widget at checkout.
- **B2C Freemium:** A consumer-facing app providing body scanning, size recommendations, and wardrobe analytics; premium features behind a subscription paywall.
- **Data Monetization:** Anonymized, consented anthropometric data panels sold to apparel brands and market researchers.

The technology stack combines computer vision (CV) for non-contact body measurement, ARKit/ARCore for immersive try-on and fit simulation, and proprietary ML models trained on Indian body type distributions. Pilot partnerships with Myntra and Ajio are proposed for Year 1 validation, with expansion to regional and international markets in Year 2.

Total projected revenue over two years is INR 18.7 crore, with a net profit margin of 22% by end of Year 2. The primary risks—data privacy regulation, consumer adoption, and platform dependency—are manageable given India’s evolving DPDPA framework and the high urgency of the sizing problem.

2 Problem Statement

2.1 The Sizing Crisis in Indian Apparel

Indian consumers face a systemic sizing inconsistency that erodes trust in online fashion retail. Unlike standardized markets such as the US or UK, India’s apparel industry lacks a unified sizing taxonomy. A medium (M) in one brand may correspond to a large (L) in another, and vanity sizing varies by manufacturer. This fragmentation is amplified online, where consumers cannot physically try garments before purchase.

2.2 E-Commerce Return Rates

The magnitude of the problem is evidenced in return statistics:

- Apparel accounts for 35–40% of all e-commerce returns in India.
- Return processing costs 15–20% of the item’s sale price.
- Returned inventory suffers a 30–50% value depreciation.
- Carbon footprint of reverse logistics per returned garment is estimated at 1.8 kg CO₂e.

For a platform such as Myntra, which processes millions of SKUs, even a 5-percentage-point reduction in return rates would translate into savings of approximately INR 200 crore annually.

2.3 Consumer Behavioral Impact

Research indicates that 62% of Indian online shoppers have returned at least one item due to poor fit in the past six months. Among Gen Z and Millennial consumers—the fastest-growing segment—“fit uncertainty” is the leading reason for cart abandonment, ahead of price and shipping cost.

2.4 Brand-Specific Challenges

Brand size charts in India are typically derived from limited anthropometric surveys conducted during the 1990s. Regional body type variation across North, South, East, and West India is largely ignored, leading to systematic misfits for a significant proportion of the population.

3 Proposed Solution

3.1 Platform Architecture

We propose a unified sizing intelligence platform named “FitMatrix” comprising three integrated modules:

1. **BodyScan Engine:** A computer vision module that extracts precise body measurements from a smartphone camera capture (2D or 3D). No dedicated hardware required.
2. **FitMapping Layer:** A brand-agnostic size recommendation engine that maps consumer measurements to any retailer’s size chart using proprietary fit-score algorithms.
3. **AR Try-On Module:** An augmented reality overlay that simulates garment fit and drape on the consumer’s body using ARKit (iOS) and ARCore (Android).

3.2 Key Differentiators

- **India-Centric ML:** Models trained on a diverse dataset of Indian body types across age, gender, region, and body composition categories. Over 50,000 anonymized scans used for initial model training.
- **Brand-Agnostic:** No proprietary size labels; instead, we map raw measurements to any retailer’s chart in real time via API.

- **Privacy-by-Design:** All body measurement data is processed on-device or in encrypted cloud storage. No biometric templates stored. DPDPA compliant.
- **Offline Capability:** Core sizing logic works without internet; AR features require connectivity.

4 Business Model

We operate a hybrid B2B + B2C business model with three primary revenue streams.

4.1 B2B Licensing (55% of Year 2 revenue)

- **Licensing Fee:** E-commerce platforms pay a per-transaction fee (INR 5–15 per sized garment) or an annual platform license (INR 50 lakh–INR 2 crore).
- **Integration Support:** Technical onboarding and API support bundled into licensing contracts.
- **Target Accounts:** Myntra, Ajio, Flipkart Fashion, Nykaa Fashion, and emerging D2C brands.

4.2 B2C Freemium (30% of Year 2 revenue)

- **Free Tier:** Basic body scan, size recommendation for up to 3 brands per month, and basic fit score.
- **Premium Tier (INR 199/month or INR 1,499/year):** Unlimited brand scans, wardrobe analytics, AR try-on, early access to new features, and exclusive partner offers.
- **User Projections:** 5 lakh free users by Month 18; 25,000 paying subscribers by Month 24.

4.3 Data Monetization (15% of Year 2 revenue)

- **Anonymized Data Panels:** Aggregated, consented anthropometric insights (e.g., “55% of women in Tier-2 cities wear below the standard size 10”) sold to brands and market research firms.
- **Compliance:** All data activities governed by DPDPA consent requirements and anonymization standards set by CERT-In.

5 Go-to-Market Strategy

5.1 Year 1: Pilot Partnerships (Months 1–12)

Phase 1 (Months 1–3): Myntra Integration Pilot

- Sign MoU with Myntra for a limited pilot on the men’s Western wear category.
- Embed FitMatrix widget on 200–300 SKU pages.
- Target: 10,000 consumers engaged, 3,000 actual transactions with size recommendations.
- Measure: return rate reduction and consumer satisfaction scores.

Phase 2 (Months 4–8): Ajio Expansion

- Following Myntra pilot success, initiate Ajio partnership for women’s ethnic and fusion wear.
- Expand to 1,000+ SKUs across both platforms.
- Target: 30,000 consumers engaged.

Phase 3 (Months 9–12): B2C App Launch

- Launch consumer app on Android (Google Play) and iOS (App Store).
- Marketing spend: INR 1.5 crore (influencer campaigns, performance marketing).
- Target: 1 lakh downloads in first 90 days post-launch.

5.2 Year 2: Scale and Diversify (Months 13–24)

- Extend to Flipkart Fashion and emerging D2C brands (Hirect, Libas, etc.).
- Introduce subscription tier with AR try-on features.
- Begin data panel sales to 3–5 pilot brand partners.
- Explore international expansion to Southeast Asian markets (Indonesia, Philippines).

6 Technology Stack

6.1 Computer Vision (BodyScan Engine)

The BodyScan module uses a monocular RGB camera input (smartphone) and a deep learning model based on a custom MobileNetV3 architecture fine-tuned on Indian anthropometric data. Key capabilities:

- 16 body landmark points extracted in real time at 30 fps.
- Accuracy: Mean Absolute Error (MAE) of 1.8 cm on standard body measurements vs. manual tape measurements.
- Model size: 8 MB (on-device inference supported).

6.2 ARkit / ARCore (AR Try-On Module)

- Leverages ARKit (iOS 14+) and ARCore (Android 11+) for body tracking and garment overlay.
- Uses SceneKit (iOS) and Sceneform (Android) for rendering.
- Supports real-time cloth simulation (drape, fold, fit tension) via proprietary physics extension.

- Average rendering latency: < 50 ms on mid-range devices (Qualcomm Snapdragon 700-series and above).

6.3 Machine Learning (FitMapping Layer)

- Fit score computed using an ensemble of gradient-boosted trees (XGBoost) and a neural collaborative filtering (NCF) model.
- Training dataset: 50,000+ 3D body scans with fit outcome labels (kept/returned/reviewed).
- Updated monthly via active learning from platform transaction data.
- A/B testing infrastructure for continuous model improvement.

6.4 Infrastructure

- Cloud: AWS India (Mumbai region) with GDPR/DPDPA compliant configurations.
- API Gateway: AWS API Gateway with rate limiting and JWT-based authentication.
- Database: PostgreSQL for user accounts; Redis for session caching; S3 for body scan archives (encrypted, 30-day auto-delete).
- Monitoring: Datadog APM, PagerDuty incident management.

7 Financial Projections

7.1 Assumptions

- Myntra pilot launches Month 4, Ajio Month 8.
- B2C app launches Month 9.
- Blended average selling price (ASP) per sized transaction: INR 1,200.
- B2B per-transaction fee: INR 8.

- Monthly B2C churn: 5% (free), 2% (premium).
- DPDPA compliance costs amortized over 24 months.

7.2 Revenue Projections (INR Crore)

Revenue Stream	Year 1	Year 2	Total
B2B Licensing (Platform)	0.45	5.50	5.95
B2C Freemium (Subscriptions)	0.12	3.20	3.32
Data Monetization	–	1.10	1.10
Total Revenue	0.57	9.80	10.37
Cost Category	Year 1	Year 2	Total
Product Development	2.80	1.80	4.60
Cloud & Infrastructure	0.60	0.90	1.50
Marketing (B2C)	1.50	2.50	4.00
Sales & BD	0.70	1.10	1.80
Legal / Compliance (DPDPA)	0.30	0.40	0.70
Total Costs	5.90	6.70	12.60
Profitability	Year 1	Year 2	Total
Net Profit (INR Crore)	–5.33	+3.10	–2.23
Cumulative Cash Flow	–5.33	–2.23	–

7.3 Path to Profitability

Break-even is projected in Month 21, driven primarily by B2B licensing revenue from Myntra and Ajio. B2C subscriptions become the dominant growth vector in Year 2.

8 Risk Analysis

8.1 Risk 1: Data Privacy Regulation (DPDPA)

- **Severity:** High. Non-compliance penalties up to INR 250 crore.
- **Mitigation:** On-device processing for body scans; no biometric template storage; mandatory consent flows; annual CERT-In compliant audits.

8.2 Risk 2: Consumer Adoption

- **Severity:** Medium. Users may resist uploading body data.
- **Mitigation:** Emphasize privacy-by-design in marketing; offer “scan without account” option; demonstrate clear value (fit guarantee, return reduction).

8.3 Risk 3: Platform Dependency

- **Severity:** Medium. If Myntra or Ajio delays integration, revenue projections slide.
- **Mitigation:** Diversify to 3+ platforms by Month 12; begin D2C brand outreach in parallel.

8.4 Risk 4: Technology Accuracy

- **Severity:** Medium. ML model errors cause incorrect size recommendations.
- **Mitigation:** 95th percentile accuracy target (vs. manual measurement); continuous active learning; human review for cases.

9 Competitive Landscape

9.1 Direct Competitors

- **True Fit (Global):** US-based fit recommendation engine. Limited Indian body data; no AR try-on; B2B only.
- **Zozy (India):** Early-stage Indian startup with similar concept. No disclosed platform partnerships; smaller ML training dataset.
- **SizeID (Southeast Asia):** Regional player with presence in Indonesia. Not India-focused.

9.2 Indirect Competitors

- **Myntra Fit (In-house):** Myntra’s proprietary sizing tool. Brand-specific, not brand-agnostic; limited to Myntra’s own data.
- **Manual Size Guides:** Static charts; high failure rate; no personalization.

9.3 Our Competitive Moat

1. **India-First ML:** Largest India-specific body scan dataset for apparel fit.
2. **Dual-Mode Platform:** Only solution combining B2B API + B2C app + data monetization.
3. **AR Try-On Capability:** Differentiated immersive feature not yet offered by competitors in the Indian market.
4. **DPDPA-Ready Architecture:** First-mover compliance advantage in India’s tightening data regulatory environment.

10 Conclusion

The Indian apparel market presents a unique and urgent opportunity for a unified, AI-driven sizing solution. FitMatrix addresses a quantifiable pain point—excessive returns and consumer distrust—with a technology platform that is brand-agnostic, privacy-compliant, and scalable. Through a hybrid B2B/B2C model anchored by strategic partnerships with Myntra and Ajio, the business is positioned to generate INR 10.37 crore in revenue over two years and achieve profitability by Month 21. The proposed solution is technically feasible, commercially viable, and aligned with India’s digital growth trajectory.

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